COLUMBIA | ENGINEERING
The Fu Foundation School of Engineering and Applied Science

Project

COURSE: ENME E6220 – Stochastic Engineering Mechanics

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1. Governing equations

Linear problem. Consider first the linear oscillator

$$m\ddot{x} + c\dot{x} + kx = w(t),$$

with initial conditions $x(0) = 0$, $\dot{x}(0) = 0$, and parameter values $m = 1$, $c = 0.5$, $k = 1$, $S_0 = 0.05$. Here $w(t)$ is Gaussian white noise of intensity S_0.

Nonlinear Duffing problem. Consider next the Duffing oscillator with combined stochastic and deterministic harmonic forcing

$$m\ddot{x} + c\dot{x} + kx + \epsilon x^3 = w(t) + F\cos(\omega t),$$

with initial conditions $x(0) = 0$, $\dot{x}(0) = 0$, and parameter values $m = 1$, $c = 0.5$, $k = 1$, $\epsilon = 1$, $F = 1.5$, $\omega = 2$, $S_0 = 0.05$. The main response quantity of interest is the joint transition PDF $p(x_f, \dot{x}_f, t_f \mid 0,0,0)$, to be evaluated at the three time instants $t_f = 0.5$, 1, and 2 s.

2. Required tasks

Q1. Perform Monte Carlo simulation for the linear oscillator using 10,000–50,000 realizations. Compute and plot the sample mean and variance histories of both $x(t)$ and $\dot{x}(t)$. Use the variance histories to assess whether the process has effectively reached stationarity. At selected times, estimate and plot the joint PDF in the $(x, \dot{x})$ plane, together with the corresponding marginals $p(x)$ and $p(\dot{x})$.

Q2. For the same linear oscillator, implement the WPI solution treatment numerically on a terminal-state grid and also implement the corresponding analytical linear benchmark in code. Compare the Monte Carlo results, the numerical WPI results, and the analytical WPI results by showing both joint PDFs and marginals.

Q3. Perform Monte Carlo simulation for the nonlinear Duffing oscillator using 10,000–50,000 realizations. Clearly state the time integration scheme used. Estimate and plot the joint PDF $p(x_f, \dot{x}_f, t_f \mid 0,0,0)$ at $t_f =$ 0.5, 1, and 2 s, and also plot the corresponding marginals $p(x_f)$ and $p(\dot{x}_f)$.

Q4. Implement the brute-force WPI solution treatment for the nonlinear Duffing oscillator on a dense terminal-state grid of 51×51 points in the $(x_f, \dot{x}_f)$ plane, for each of the three time instants. This dense-grid WPI solution will serve as the deterministic benchmark for the extrapolation and interpolation studies. Show the corresponding joint PDFs and marginals.

Q5. Repeat the nonlinear WPI computation on several coarse terminal grids, for example $N = 5$, 7, 9, and 11 points per stochastic dimension. The purpose of this stage is to determine how coarse the original WPI grid may be while still allowing satisfactory reconstruction of the dense-grid PDF after extrapolation.

Q6. For each coarse WPI grid, extrapolate the corresponding sparse-grid WPI solution to a dense 51×51 grid. Compare the extrapolated results against the brute-force WPI solution obtained directly on the same dense grid. The comparison must be based on the visual agreement of the joint PDFs and marginals. Based on these comparisons, determine the smallest number of coarse-grid points per dimension that still reproduces the dense-grid WPI solution satisfactorily.

Q7. Using the same coarse-grid WPI data, compare the WPI-based extrapolation approach against 2 to 3 purely numerical interpolation methods. Examples include bilinear interpolation, bicubic interpolation, spline interpolation, and radial basis function interpolation. Compare the resulting dense-grid PDFs against the brute-force WPI solution on the 51×51 grid by showing both joint PDFs and marginals.

Q8. Using the smallest coarse WPI grid found acceptable in Q6, extrapolate the nonlinear WPI solution to a much denser 151×151 grid, and compare this final result against Monte Carlo. Present both joint PDFs and marginals.

3. General instructions

- In every WPI-based figure, show explicitly the original WPI terminal-grid points, so that it is clear which information comes directly from WPI and which comes from extrapolation or interpolation.
- The report must include both **joint PDFs** and **marginal PDFs** throughout. For the joint PDFs, contour plots or surface plots in the $(x_f, \dot{x}_f)$ plane may be used. For the marginals, both $p(x_f)$ and $p(\dot{x}_f)$ must be shown.
- Formal numerical error metrics are not required. The assessment should rely primarily on the visual agreement of the joint PDFs and marginals.
- Report runtime in seconds and speed-up ratio. State clearly how runtime is measured. At a minimum, the speed-up ratio must be reported for the following comparisons:
  1. brute-force WPI on 51×51 versus coarse WPI plus extrapolation to 51×51;
  2. Monte Carlo versus coarse WPI plus extrapolation to 151×151.

4. Deliverables

A simple course-report format is sufficient. The final report should be organized as follows: problem statement; Monte Carlo methodology; linear validation study; brute-force nonlinear WPI benchmark; coarse-grid extrapolation study; interpolation comparison; runtime and speed-up results; discussion and conclusions. Include also the code used (Matlab or Python).